Creating an Experiment Scenario for AIM

Why This Research

Life experiences shape how we think, feel, and make decisions. But we only ever live one life. This means people may support a different set of policies if they had different experiences. The goal of the As-If Machine is to simulate how a participant's life would change under various counterfactuals, and we test if these personalized simulations change perceived psychological proximity as well as policy support.

As a concrete example, to increase support for science funding, we had the As-If Machine imagine how cuts to science funding would affect a participant's own life. We then found that co-writing with our system decreased psychological proximity to science funding and (in some cases) increased support for science funding.

How AIM Works

The diagram below shows the full system pipeline — from intake form to autocomplete.

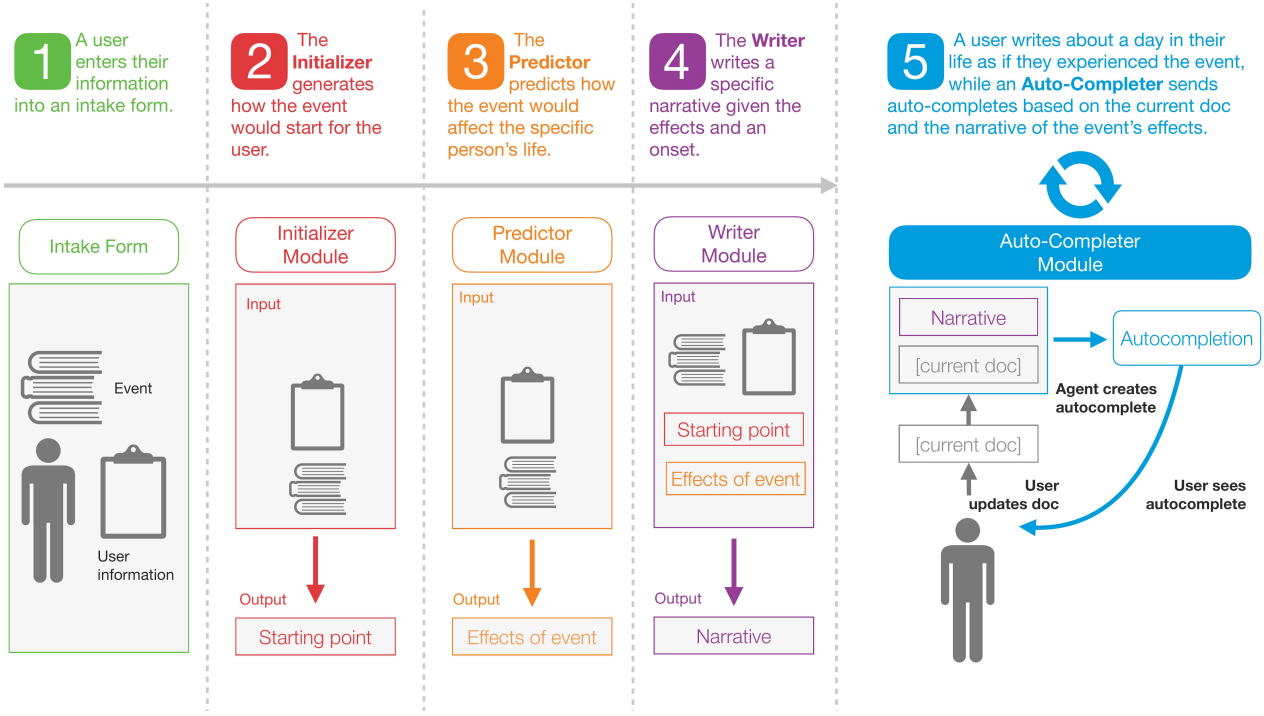


Figure 1. AIM system overview.

What Is AIM?

AIM is a multi-agent writing system where the user-facing part completes sentences in real time as participants type a short first-person story. The AI does not use a single fixed prompt — it assembles a new prompt on every keypress from several sources: the scenario the researcher programs, the answers a participant gave on an intake form, a realistic plot that agents synthesized, and what the user wrote so far.

Because the AI reads the participant's intake answers in every call, questions that capture personally meaningful details — finances, relationships, routines — make the suggestions feel tailored rather than generic.

Autocomplete Prompt Anatomy

The diagram below shows the structure of a single API call. **CHARACTER** is where intake form answers go — that block is rebuilt fresh for every participant, every keypress.

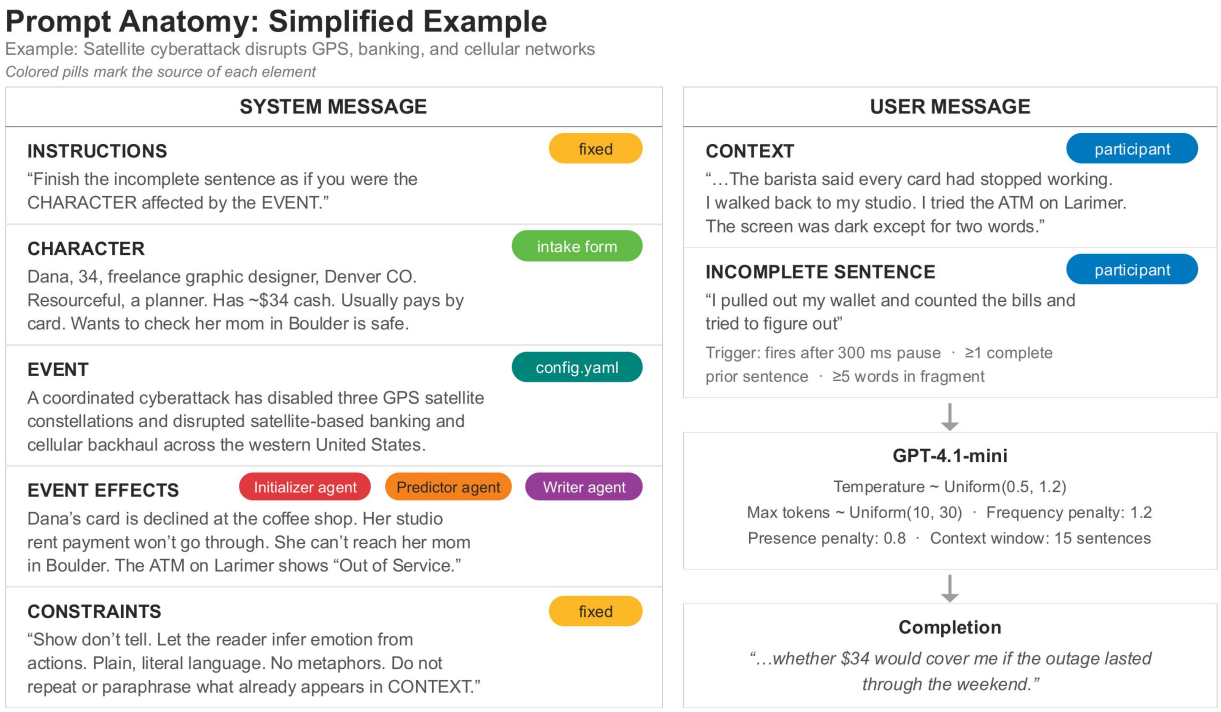


Figure 2. Structure of one autocomplete call. Green pill = intake form. Teal = config.yaml. Red/orange/purple = agent pipeline. Blue = participant's writing. Yellow = fixed.

Your Task

You will create one scenario by editing a configuration file. A scenario has two parts: (1) a **situation** the participant imagines happening to them, and (2) an **intake form** that collects personal details used to tailor the AI suggestions.

Step 1 — See How the YAML Fields Map to the Interface

The configuration file controls everything the participant sees. The diagram below shows how each field maps to the editor.

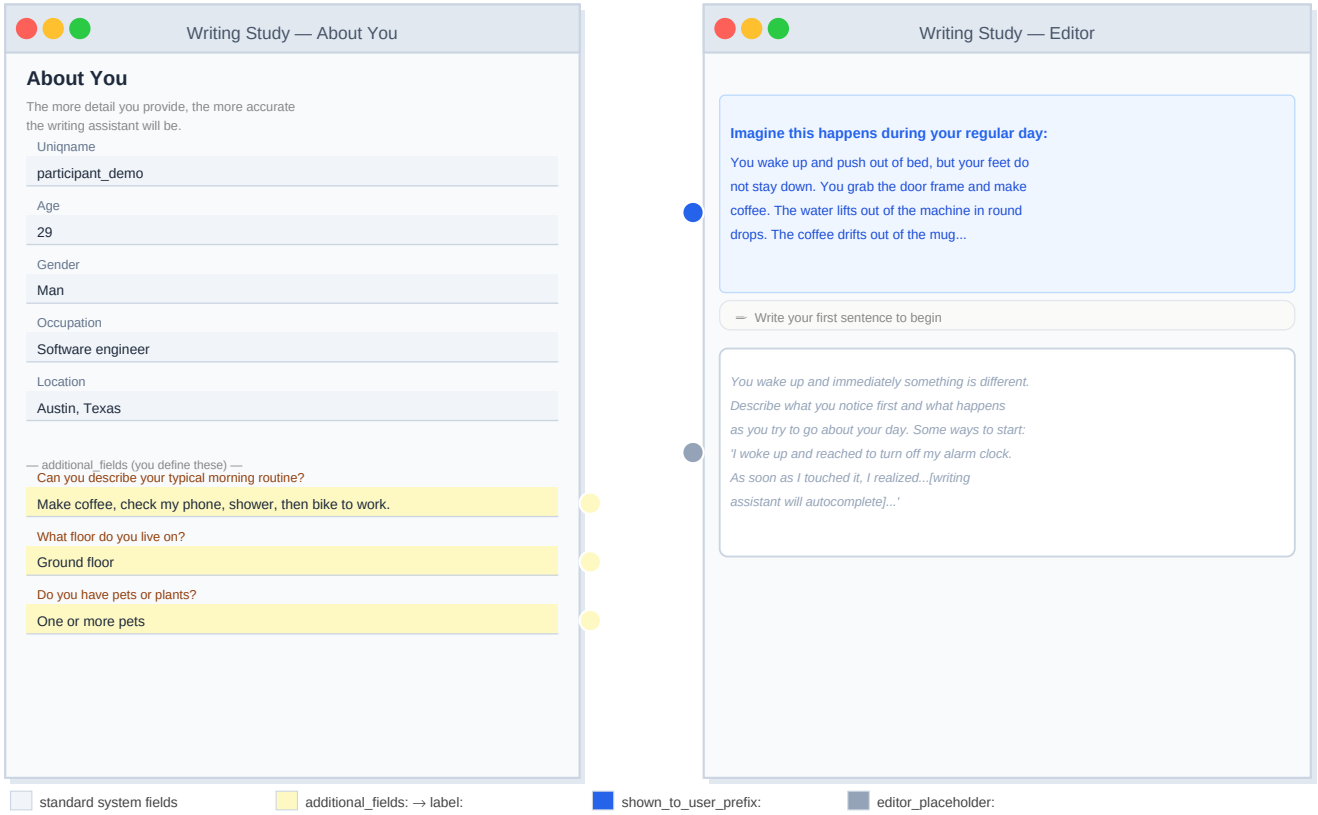


Figure 3. Left: intake form (additional_fields). Right: writing editor. Yellow = additional_fields questions you define. Blue = shown_to_user_prefix. Gray italic = editor_placeholder.

Step 2 — Study a Worked Example

Suppose the scenario is: *a participant's building has just been sold and they must vacate in 60 days.* A vague scenario produces vague completions. A specific one — with concrete numbers and a [Tip:] to guide the AI — produces writing that feels real.

Making the scenario specific

Rather than writing “your landlord sold the building,” let's make the scenario more specific and add concrete pressure:

- *"You must vacate in 60 days."*
- *"Comparable apartments in your area cost $400 more per month than you currently pay."*

Then add a [Tip: ...] to tell the AI *how* to write — not what happens, but how to render it:

- *[Tip: Show the concrete search process — browsing listings, calling landlords, doing the math. Do not name emotions; show them through actions and specific details.]*

Personalizing with intake questions

To tailor the AI's suggestions to each participant, collect details that would plausibly change the story. For a housing scenario, useful questions are:

- What is your current housing situation?
- How much of your monthly income goes to rent?
- What would make finding new housing hardest for you?

These answers feed directly into the AI's prompt on every keypress — so a participant who rents alone and spends 40% of income on rent gets different suggestions than one who owns their home.

The YAML that produces this scenario

Here is the complete block. Study it! Then you will write your own.

```
# ■■ Give your scenario a unique snake_case key (no spaces) ■■■■■■■■■■■■■■■■■■■■■
  event_mine:

    # THE SCENARIO DESCRIPTION (backend only — not shown directly to participants)
    # Used to generate the story plot that appears above the editor.
    # Write in second person ("you"). Be specific — vague scenarios produce
    # generic completions. The [Tip: ...] guides the AI; it is never shown.
    name: >
      Your landlord informs you that your building has been sold. You must
      vacate in 60 days. Comparable apartments in your area cost $400 more
      per month than you currently pay.
      [Tip: Show the concrete search process — browsing listings, calling
      landlords, doing the math. Do not name emotions; show them through
      actions and specific details.]

    # TEXT SHOWN ABOVE THE EDITOR (the bold lead-in before the story plot)
    shown_to_user_prefix: "Imagine this happens"

    # PLACEHOLDER TEXT inside the writing editor
    editor_placeholder: "You just got the 60-day notice. Start writing
      about what you do next. Press Tab to accept AI suggestions."

    # ■■ LEAVE THESE THREE LINES EXACTLY AS-IS ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    autocomplete_context: "plot_from_effects"
    shown_to_user:        "start"
    effects:

    # THE INTAKE FORM
    # Each entry becomes one question on the "About You" page.
    # Answers feed directly into the CHARACTER block of every autocomplete call.
    # Design 3–5 questions. Best questions capture variation that would change
    # the story: finances, routines, relationships, constraints.
    additional_fields:

      # DROPDOWN — participant picks one option from a list
      housing_situation:                      # unique snake_case key
        type:    "SelectField"
        label:   "What is your current housing situation?"   # question shown
        choices:
          - "Renting alone"
          - "Renting with roommates"
          - "Own my home"
          - "Living with family"
        validators: [ "DataRequired" ]          # leave as-is

      # DROPDOWN
      monthly_rent:
        type:    "SelectField"
        label:   "How much of your monthly income goes to rent?"
        choices:
          - "Less than 25%"
          - "25–40%"
          - "More than 40%"
        validators: [ "DataRequired" ]

      # FREE-TEXT — participant types their own answer
      biggest_obstacle:
        type:    "StringField"
        label:   "What would make finding new housing hardest for you?"
        validators: [ "DataRequired" ]
```

Field type	Use when…

SelectField + choices:	Participant picks one option from a dropdown list
StringField	Participant types their own answer (free text)

Step 3 — Create Your Own Scenario

Now design your own scenario. It can be about anything, really. Think about what kind of experience, if a person had it, would shift how they think or feel — and what details would make it feel specific to *them.* You can also see some examples below if you want to pick one of these.

Some directions to consider

Direction	Examples
Imagined worlds (reality works differently)	Sleep has been eliminated biologically — permanently — and it is the first year of a world that gained eight waking hours a day and has no idea what to do with them. Land can no longer be owned; you pay for time in a place rather than title to it, and homes rotate. Facial expressions are now fully voluntary — no face betrays anything its owner hasn't decided to show.
Empathy-promoting experiences (a body or life unlike your own)	You suddenly develop trouble hearing. You develop chronic pain that makes every ordinary task negotiable. You develop face blindness and can no longer recognize anyone — including people you love — by their face alone. You develop a severe stutter that appears only under stress, which means it appears in every situation that matters.
Ordinary life, derailed (a mundane system fails you)	Your landlord sells the building and you have to move. Your car breaks down and, due to right-to-repair restrictions, the supplier can't get the part for three months. Your doctor retires; the practice is acquired by a health system not accepting new patients, and you have a condition that needs monitoring. Your bank is acquired mid-month; your direct deposit is delayed two weeks, right before rent is due, and every rep says it will resolve itself.

Design your scenario — think through these questions

1. What is the core experience? What happens, specifically?
2. What concrete detail would make it feel real (a number, a deadline, a physical sensation)?
3. What would the AI need to know about the participant to write suggestions that feel personal to them?

Then edit config.yaml

Now go to the application window. The config.yaml file is already open. You will use the housing example above as your starting point and replace the content with your scenario. Use a unique `snake_case_key` for your event.

Step 4 — Run It

Once you have saved your changes to `config.yaml`, start the server:

```
python3 app.py
```

Then open your scenario in the browser:

```
http://localhost:5001/user_settings_experiment/your_key_here

# Replace  your_key_here  with the snake_case key you used in config.yaml
```

Every time you change config.yaml, stop the server (Ctrl+C) and rerun `python3 app.py`.

When you are done, return to the Qualtrics survey to complete the post-task questions.